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(20516) Roll No. .....

BCA-IV Sem.

18019

B. C. A. Examination, May 2016

OPTIMIZATION TECHNIQUES

(BCA-404)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A
(Very Short Answer Questions)

Attempt all the *five* questions of this Section. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. Define a general and standard linear programming problem.

2. Solve the following LPP by graphical method :

Maximize : $z = 2x_1 + x_2$

Subject to : $3x_1 + 4x_2 \le 6$

$6x_1 + x_2 \le 3$

$x_1, x_2 \ge 0.$

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3. Customers arrive at a booking office window, being manned by a single individual at a rate of 25 per hour. Time required to serve a customer has exponential distribution with a mean of 120 seconds. Find the mean waiting time of a customer in the queue.

4. Draw economic order quantity graph showing the relationship of inventory costs with order quantity and inventory level overtime.

5. Explain briefly replacement policies for items whose efficiency deteriorates with time.

Section-B
(Short Answer Questions)

Attempt any *two* questions from this Section. Each question carries 7½ marks. 7½×2=15

6. Find the sequence that minimizes total elapsed time to complete the following six jobs and also find the minimum time :

Jobs :	1	2	3	4	5	6
MachineI :	3	12	15	6	10	9
MachineII :	8	10	10	6	12	3

7. Solve the following assignment problem represented by the matrix :

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	I	II	III	IV	V
A	6	5	8	11	16
B	1	13	16	1	10
C	16	11	8	8	8
D	9	14	12	10	16
E	10	13	11	8	16

8. Obtain the steady state equations for the model {(M/M/1):(∞/FCFS)} and also find the formula for mean and the variance of the queue length.

Section-C

(Detailed Answer Questions)

Attempt any *three* questions from this Section. Each question carries 15 marks. 15×3=45

9. Determine an optimum basic feasible solution to the transportation problem given below :

	D_1	D_2	D_3	D_4	
O_1	1	2	3	4	6
O_2	4	3	2	0	8 Capacity
O_3	0	2	2	1	10
	4	6	8	6	24
	Demand				

where O_i and D_j denote ith origin and jth destination respectively.

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10. Use simplex method to solve the following LPP :

Maximize : $Z = 4x_1 + 10x_2$

Subject to : $2x_1 + x_2 \le 50$

$2x_1 + 5x_2 \le 100$

$2x_1 + 3x_2 \le 90$

$x_1, x_2 \ge 0$

11. Obtain the dual problem of the following LPP :

Maximize : $f(x) = 2x_1 + 5x_2 + 6x_3$

Subject to : $5x_1 + 6x_2 - x_3 \le 6$

$-2x_1 + x_2 + 4x_3 \le 4$

$x_1 - 5x_2 + 3x_3 \le 1$

$-3x_1 - 3x_2 + 7x_3 \le 6$

$x_1, x_2, x_3 \ge 0$

Also verify that the dual of the dual problem is the primal problem.

12. Derive the Wilson EOQ formula. What are the practical limitations of EOQ formula? Also discuss the costs involved in an inventory problem.

13. Explain the following :

(i) Present worth factor (pwf)

(ii) Discount rate

(iii) Dual simplex method

(iv) Group replacement and individual replacement policy

(v) Tic-tac problem.

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N (Printed Pages 7)

(20517) Roll No.....

BCA-IV Sem.

18019

B.C.A. Examination, May 2017

Optimization Techniques

(BCA-404)

(New)

Time : Three Hours *Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

Note : Attempt **all five** questions. Each question carries 3 marks.

1. What is the condition in algorithm that an L.P.P. has unbounded solution?

2. Define inventory problem. List the various

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costs associated with an inventory problem.

3. Define money value, present value and discount rate.

4. Define queue length, waiting time and busy period.

5. Explain sequencing problem.

Section-B

Note : Attempt any two questions from this section. Each question carries 7.5 marks.

$7\frac{1}{2} \times 2 = 15$

6. Solve the following assignment.

Problem :

Man → / Job ↓	1	2	3	4
I	12	30	21	15
II	18	33	9	31
III	44	25	24	21
IV	23	30	28	14

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7. Solve by graphical method the linear programming problem .

$$\min z = 20x_1 + 10x_2$$

such that $x_1 + 2x_2 \le 40$

$3x_1 + x_2 \ge 30$

$4x_1 + 3x_2 \ge 60$

and $x_1, x_2 \ge 0$

8. We have five jobs, each of which have to go through the machines A, B and C in the order ABC.

Processing time in hours

Jobs	1	2	3	4	5
Machine A (A_i)	5	7	6	9	5
Machine B (B_i)	2	1	4	5	3
Machine C (C_i)	3	7	5	6	7

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Determine a sequence for the jobs that will minimize the total elapsed time.

Section-C

Note : Attempt any **three** questions out of the following **five** questions. Each question carries 15 marks.

9. Solve the following LPP

$$\text{Max } z = 40x_1 + 35x_2$$

subject to

$2x_1 + 3x_2 \le 60$

$4x_1 + 3x_2 \le 96$

$x_1, x_2 \ge 0$

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10. Solve the following transportation problem

	D_1	D_2	D_3	Available
O_1	2	7	4	5
O_2	3	3	1	8
O_3	5	4	7	7
O_4	1	6	2	14
Required	7	9	18	

11. On average 96 patients per 24 hour day require the service of an emergency clinic. Also on average a patient requires 10 minutes of active attention. Assume that the facility can handle only one emergency at a time. Suppose that it costs the clinic Rs. 100 per patient treated to obtain an average serving time of 10 patients and that each minute of of decrease in the average time would cost by Rs. 10 per patient treated. How much

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would to be budgeted by the clinic to decrease the average size of the queue from one and one third patients to half a patient?

12. The cost of new machine is Rs. 5000. The maintenance cost of nth year is given by $C_n=500(n-1)$, n=1, 2 -----. Suppose that the discount rate per year is 0.5. After how many years it will be economical to replace the machine by new one?

13. Use graphical method to minimize the time needed to process the following jobs on machines shown below i.e. for each machine find the job which should be done first. Also calculate the total time needed to complete both jobs.

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Job1		Job2	
Sequence of Machines	Time	Sequence of Machines	Time
A	2	C	4
B	3	A	5
C	4	D	3
D	6	E	2
E	2	B	6

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(20518) Roll No. .....

BCA- IV Sem.

18019

B. C. A. Examination, May 2018

Optimization Techniques

(BCA-404)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. Define a linear programming problem.

2. Define money value and present value.

3. Define busy period, idle period and mean arrival rate.

4. Explain travelling salesman problem.

5. Describe holding cost, shortage cost and ordering cost.

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Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. Solve the following assignment problem :

Subordinates

Tasks	I	II	III	IV
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

7. The cost of an item is 3,000. The salvage value and running cost are given below. Find the most economical replacement age of the item :

Year	Running cost	Salvage cost
1	600	2000
2	700	1333
3	800	1000
4	900	750
5	1000	500
6	1200	300
7	1500	300

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8. Find the sequence that minimizes the total elapsed time required to complete the following tasks on two machine :

	A	B	C	D	E	F	G	H	I
Macine - I	2	5	4	9	6	8	7	5	4
Machine - II	6	8	7	4	3	9	3	8	11

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Solve the following L. P. P. :

Maximize $z = 5x_1 + 3x_2$

s. t. $3x_1 + 5x_2 \le 15$

$5x_1 + 2x_2 \le 10$

$x_1,\ x_2 \ge 0$

10. Solve the following transportation problem :

		To 1	2	3	Supply
From	1	2	7	4	5
	2	3	3	1	8
	3	5	4	7	7
	4	1	6	2	14
	Demand	7	9	18	34

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11. The cost pattern for two machines A and B when money value is not considered is given as follows :

Year	Cost at the beginninng of year in Rs.	
	Machine A	Machine B
1	900	1400
2	600	100
3	700	700
Total	2200	2200

Find the cost pattern for each machine when money worth is 10% per year and hence find which machine is less costly.

12. We have five jobs, each of which must go through the machine A, B and C in the order ABC :

Processing Times in hours

Job No. i	1	2	3	4	5
Machine A(A_i)	5	7	6	9	5
Machine B (B_i)	2	1	4	5	3
Machine C(C_i)	3	7	5	6	7

Determine a sequence for the jobs that will minimize the total elapsed time.

13. Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate at 20 per hour. The time required to serve a customer has an expontential distribution with a mean of 100 seconds. Find the average waiting time of a customer and queue length.

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(20519) Roll No. .....

Total Questions : 13] [Printed Pages : 7

18019

B.C.A. IVth Semester Examination, May-2019

OPTIMIZATION TECHNIQUES

(BCA–404)

Time : 3 Hrs.] *[M.M. : 75*

Note :– Attempt all the Sections as per instructions.

Section–A

(Very Short Answer Type Questions) 3×5=15

Note :– Attempt all the *five* questions. Each question carries 3 marks.

1. Write a linear programming problem in matrix form.
2. Describe classification of inventory models.
3. Describe present value and discount rate.

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4. Explain sequencing problem.
5. Explain queue length, waiting time and traffic intensity.

Section–B

(Short Answer Type Questions) 7½×2=15

Note :– Attempt any *two* questions from this section. Each question carries 7½ marks.

6. Solve the following assignment problem :

Person

		A	B	C
Job	1	120	100	80
	2	70	90	110
	3	110	140	120

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7. The cost of a machine is Rs. 6,100 and its resale value is only Rs. 100. The maintenance costs are found from experience to be as under :

Year	1	2	3	4	5	6	7	8
Maintenance Cost in Rs.	100	250	400	600	900	1250	1600	2000

When should machine be replaced.

8. We have fine jobs each of which must to through two machines A and B in the order AB. Processing times in hours are given below :

Job	1	2	3	4	5
Machine A (A_i)	5	1	9	3	10
Machine B (B_i)	2	6	7	8	4

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Determine sequence for the fine jobs that will minimize the elapsed time T.

Section–C

(Long Answer Type Questions) 3×15=45

Note :– Attempt any *three* questions out of the following *five* questions. Each question carries 15 marks.

9. Solve the following LPP.

$$\max. Z = 2x_1 + 4x_2$$

$$\text{s.t.} \quad 2x_1 + 3x_2 \le 48$$

$$x_1 + 3x_2 \le 42$$

$$x_1 + x_2 \le 21$$

$$x_1, x_2 \ge 0$$

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10. Solve the following transporation problem :

		To				
		1	2	3	4	Supply
From	1	3	6	8	5	20
	2	6	1	2	5	28
	3	7	8	3	9	17
Demand		15	19	13	18	

11. Let the value of money be assumed to be 10% per year and suppose that machine A is replaced after every 3 years whereas machine B is replaced after every six years. The yearly costs of the machines are given as under :

Year	1	2	3	4	5	6
Machine A	1000	200	400	1000	200	400
Machine B	1700	100	200	300	400	500

Determine which machine should be purchased.

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12. We have five jobs each of which must go through the machines A, B and C in order ABC. Processing times are :

Job	A	B	C
1	4	5	8
2	9	6	10
3	8	2	6
4	6	3	7
5	5	4	11

Determine a sequence for the fine jobs that will minimize the elapsed time.

13. ATV repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which

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they came in and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day. What is repairman's expected idle time each day ? How many jobs are ahead of the average set just brought ?

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(20622) Roll No. ....

BCA-IV Sem.

18019

B.C.A. Examination, June-2022

OPTIMIZATION TECHNIQUES

(BCA-404)

Time : Three Hours | *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt **all** the **five** questions. Each question carries 3 marks.

1. Define optimal solution of a linear programming problem.

2. Define deterministic model in inventory theory.

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3. Define sequencing problem.

4. Explain queue length, waiting time and traffic intensity.

5. Explain group replacement and individual replacement.

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions from this section. Each question carries 7½ marks.

6. Solve the following assignment problem:

Man

Job	1	2	3	4
I	12	30	31	15
II	18	33	9	31
III	44	25	24	21
IV	23	30	28	14

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7. The cost of a machine is Rs. 6100 and its scrap value is only Rs. 100. The maintenance costs are found from experience to be as below:

Year	1	2	3	4	5	6	7	8
Maintenance Cost in Rs.	100	250	400	600	900	1250	1600	2000

When should the machine be replaced.

8. we have five jobs each of which must go through two machines A and B in the order AB. Processing times in hours are given in the table below:

Job	1	2	3	4	5
Machine A	5	1	9	3	10
Machine B	2	6	7	8	4

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Determine the sequence for the five jobs that will minimize the ellipse time.

Section-C

(Long Answer Type Questions)

Note : Attempt any **three** questions out of the following **five** questions. Each question carries 15 marks.

9. Solve the following LPP:

Max. $Z=5x_1+3x_2$

s.t. $3x_1+5x_2 \leq 15$

$5x_1+2x_2 \leq 10$

$x_1 > x_2 \geq 0$

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10. Solve the following transportation problem:

		To			
		1	2	3	Supply
From	1	2	7	4	5
	2	3	3	1	8
	3	5	4	7	7
	4	1	6	2	14
Demand		7	9	16	34

11. We have five jobs, each of which must go through the machines A,B and c in the order ABC, Processing times are as follows :

	Processing times in hours		
Jobs	A	B	C
1	4	5	8
2	9	6	10
3	8	2	6
4	6	3	7
5	5	4	11

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Determine a sequence for the five jobs that will minimize the elapsed time T.

12. The cost of a new machine is Rs. 5000. The maintenance cost of nth year is given by $C_n = 500(n-1)$, $n=1,2,3,$-----. Suppose that the discount rate per year is 0.5. After how many years it will be economical to replace the machine?

13. Customers arrive at a sales counter manned by a single person according to a poissen process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution

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with a mean of 100 seconds. Find the average waiting time of a customer and queue length.

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B.C .. - IV Sem.

Printed Pages : 4

Roll No. .....

18019

B.C.A. Examination, June–2023

OPTIMIZATION TECHNIQUES

(BCA-404)

Time : 3 Hours] *[Maximum Marks : 75*

Section–A

Note: *Attempt **all** the **five** questions. Each question carries 3 marks.* 5 × 3=15

1. What is the central problem of linear programming?
2. What are the characteristics of a queuing system?
3. What is the objective of replacement theory?
4. What is a deterministic inventory model?
5. What are the different types of job sequencing problems?

Section–B

Note: *Attempt any **two** questions from this section. Each carry 7.5 marks.*

6. Explain the simplex method in linear programming, including its basic steps and how it is used to solve optimization problems.

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7. Explain the basic economic order quantity (EOQ) model and its assumptions.
8. Explain the concept of "time to failure" and "time to repair" in replacement theory.

Section–C

Note: *Attempt any **three** questions out of **five**. Each carry 15 marks.*

9. A company produces two products, A and B, which have profit margins of \$3 and \$5 per unit, respectively. The company has a limited production capacity of 1000 hours per week and a limited raw material supply of 1500 units per week. Product A requires 2 hours of production time and 1 unit of raw material, while product B requires 3 hours of production time and 2 units of raw material. Formulate the linear programming problem to maximize the total profit for the company and solve it graphically.
10. Discuss the central problem of linear programming in detail, including the various definitions, basic theorems, and properties associated with it.
11. A company uses a machine for production that has a useful life of 4 years. The cost of the machine is Rs. 20,000 and the annual maintenance cost for the machine is Rs. 2,500. The company estimates that the salvage value of the machine at the end of

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4 years will be Rs. 2,000. The company is considering two replacement policies:

(i) Replace the machine at the end of year 4;

(ii) Replace the machine as soon as it fails.

The company has collected data on the probability distribution of the machine's failure time. The probability that the machine fails in the first year is 0.1, in the second year is 0.2, in the third year is 0.3, and in the fourth year is 0.4.

(a) Calculate the expected cost for each of the two replacement policies.

(b) Which policy should the company choose?

(c) What is the probability that the machine will fail during the third year?

(d) What is the expected number of years until the machine is replaced under policy (ii)?

Note: *Assume that the company operates the machine for 4 years only.*

12. A retailer sells a particular product at a constant rate of 200 units per week. The product has a fixed ordering cost of Rs. 500 per order, and a variable holding cost of Rs. 2 per unit per week. The lead time for an order is 2 weeks, and the demand during this period is normally distributed with a mean of 400 units and a standard deviation of 50 units.

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Determine the optimal order quantity, reorder point, and the total annual inventory cost assuming 50 working weeks in a year.

13. A supermarket has two checkout counters to serve customers, The customers arrive at the checkout counters according to a Poisson process with an arrival rate of 20 customers per hour. The service time of each checkout counter follows an exponential distribution with a mean of 2.5 minutes per customer. The management of the supermarket is interested in improving the service by reducing the average waiting time in the queue.

(a) Determine the utilization rate of each checkout counter.

(b) Determine the probability of having zero customers in the system.

(c) Determine the average number of customers waiting in the queue.

(d) Determine the average waiting time in the queue,

(e) Suppose the management decides to add a third checkout counter. Determine the utilization rate of each checkout counter and the average waiting time in the queue for each customer.

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(20524) Roll No. .....

BCA-IV Sem.

18019

B.C.A. Examination, May-2024

OPTIMIZATION TECHNIQUES

(BCA-404)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt all the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all questions. Each question carries 3 marks. 5×3=15

1. Define a linear programming problem.

2. Explain deterministic model of inventory.

3. Explain characteristics of queuing system.

4. Explain money value and discount ratio.

5. Explain sequencing problem for n jobs on two machines.

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions from this section. Each question carries 7.5 marks. 2×7.5=15

6. Solve the following assignment problem:

Subordinates

		I	II	III	IV
Tasks	A	8	26	17	11
	B	13	28	4[illegible]	26
	C	38	19	18	15
	D	19	26	24	10

7. A machine owner finds from his past records that the cost per year of maintaining a machine whose purchase price is Rs. 6,000 are as given below :

Year	Maintenance Cost Rs.	Resale price Rs.
1	1000	3000
2	1200	1500
3	1400	750
4	1800	375
5	2300	200
6	2800	200
7	3400	200
8	4000	200

At what age the replacement is due?

8. We have five jobs each of which must go through two machines A and B in order AB. Processing times in hours are given in the following table :

Job	Machine A (Ai)	Machine B (Bi)
1	5	2
2	1	6
3	9	7
4	3	8
5	10	4

Determine the sequence for the five jobs that will minimize the elapsed time.

Section-C

(Long Answer Type Questions)

Note : Attempt any **three** questions out of the following **five** questions. Each question carries 15 marks. 3×15=45

9. Solve the following LPP :

Max. $z=5x_1+3x_2$

s.t. $x_1+x_2 \leq 2$

$5x_1+2x_2 \leq 10$

$3x_1+8x_2 \leq 12$

$x_1, x_2 \geq 0$

10. Solve the following transportation problem :

		To				
		1	2	3	4	Supply
From	1	1	2	1	4	30
	2	3	3	2	1	50
	3	4	2	5	9	20
Demand		20	40	30	10	100

11. Find the optimal sequence of jobs which minimize the total elapsed time based on the following informations :

Jobs	1	2	3	4	5
Machine A (Ai)	3	8	7	5	2
Machine B (Bi)	3	4	2	1	5
Machine C (Ci)	5	8	10	7	6

12. Discuss economic lot size model with uniform rate of demand, finite rate of replenishment having no shortages.

13. If you wish to have a return of 10% per annum on your investment then which of the following plans would you prefer?

	Plan A (Rs.)	Plan B (Rs.)
Ist Cost	2,00,000	2,50,000
Scrap value after 15 years	1,50,000	1,80,000
Excess of annual revenue over annual disbursement	25,000	30,000